Problem-Solution Fit Report

Project: Heart Disease Analysis using Tableau and MySQL

1. Customer Segment (CS)

Primary: Healthcare professionals such as doctors, cardiologists, and hospital administrators who analyze patient health data.

Secondary: Medical researchers and data scientists who study heart disease trends and risk factors.

Tertiary: Public health organizations and policymakers interested in improving preventive healthcare strategies.

2. Jobs-To-Be-Done / Problems (J&P)

• Analyzing large healthcare datasets related to heart disease.

• Identifying key risk factors such as age, cholesterol level, blood pressure, and lifestyle habits.

• Understanding patterns and relationships between medical attributes and heart disease occurrence.

• Converting complex raw data into meaningful visual insights.

• Supporting healthcare professionals in making data-driven decisions.

3. Triggers (TR)

• Increasing number of heart disease cases worldwide.

• Need for better data analysis tools in healthcare.

• Availability of open healthcare datasets for research and analysis.

• Demand for visual analytics tools to simplify complex medical data.

4. Emotions (EM)

Before using the system:

• Difficulty understanding large medical datasets.
• Confusion when identifying disease risk factors.
• Frustration due to time-consuming manual analysis.

After using the system:

• Better understanding of health risk factors.
• Confidence in analyzing medical data visually.
• Faster decision making through interactive dashboards.

5. Available Solutions (AS)

• **Manual Spreadsheet Analysis (Excel)**
Pros: Easy to use and widely available.
Cons: Limited visualization capability and difficult for large datasets.

• **Traditional Medical Reports**
Pros: Reliable medical information.
Cons: Static data and no interactive analysis.

• **Our Solution (Tableau Dashboard)**
Pros: Interactive visualizations, easy exploration of data, fast analysis, and better understanding of health patterns.

6. Customer Constraints (CC)

• Limited technical knowledge among healthcare staff regarding data analytics tools.

• Budget limitations for advanced healthcare analytics software.

• Need for simple and easy-to-understand dashboards for quick analysis.

7. Customer Behaviour (BE)

This section describes how users currently analyze healthcare data.

• **Manual Data Analysis:**
Medical researchers often analyze health data manually using spreadsheets.

• **Basic Charts:**
Doctors and researchers create simple charts and tables to observe patterns.

• **Limited Visualization Tools:**
Many organizations lack advanced visualization platforms to explore medical data interactively.

• **Static Reports:**
Health insights are usually shared through static reports or documents without interactive capabilities.

8. Channels of Behaviour (CH)

8.1 Online Channels

• Healthcare data portals where medical datasets are available.

• Research platforms used by medical researchers.

• Web-based dashboards and analytics tools.

8.2 Offline Channels

• Hospital meetings discussing patient health statistics.

• University laboratories where students analyze medical datasets.

• Medical research conferences and presentations.

9. Problem Root Cause (RC)

The main reasons behind this problem include:

• **Large and Complex Healthcare Data:**
Medical datasets contain numerous attributes that are difficult to analyze manually.

• **Lack of Visualization Tools:**
Many healthcare institutions rely on traditional data analysis methods.

• **Data Interpretation Challenges:**
Understanding relationships between multiple health parameters is difficult without proper visualization.

• **Growing Health Concerns:**
Increasing heart disease cases require more effective data analysis solutions.

10. Your Solution (SL)

The proposed solution provides an integrated system for analyzing heart disease data.

• **Interactive Tableau Dashboard:**
Visualizes healthcare data using charts, graphs, and filters to explore relationships between different medical attributes.

• **MySQL Database Integration:**
Stores and manages healthcare datasets efficiently for analysis.

• **Data Visualization:**
Transforms raw medical data into meaningful insights that are easy to understand.

• **Web Integration:**
Allows dashboards to be published online so users can access them through a web browser.

• **Pattern Identification:**
Helps identify important risk factors related to heart disease.